Q1. Explain the basic principles of symmetric cryptography. Describe the roles of plaintext, ciphertext, secret key, encryption algorithm and decryption algorithm, and explain Kerckhoffs's principle. [10 Marks]

Meaning: Symmetric cryptography is an encryption method in which the sender and receiver use the same secret key, or closely related secret keys, to encrypt and decrypt information. The key must be shared securely before confidential communication can take place.

Main components

• **Plaintext:** The original readable message or data that needs protection. Example: “MEET AT 5”.

• **Ciphertext:** The scrambled, unreadable form produced after encryption. It should not reveal the original message to an unauthorised person.

• **Secret key:** A value known only to authorised parties. The encryption and decryption processes depend on this key.

• **Encryption algorithm:** The mathematical procedure that converts plaintext into ciphertext using the secret key.

• **Decryption algorithm:** The reverse procedure that uses the appropriate secret key to recover the original plaintext.

Working of a symmetric cryptosystem

Plaintext (original message)	Encryption algorithm + secret key	Ciphertext (encrypted message)

The receiver applies the decryption algorithm with the same secret key to the ciphertext and obtains the original plaintext.

Ciphertext	Decryption algorithm + same key	Recovered plaintext

In mathematical form: **C = E(K, P)**, where P is plaintext, K is the secret key, E is encryption and C is ciphertext. Decryption gives **P = D(K, C)**.

Kerckhoffs's principle

Kerckhoffs's principle states that a cryptographic system should remain secure even if everything about the system, except the secret key, is publicly known. In other words, security must depend on keeping the key secret, not on hiding the algorithm. This allows algorithms to be openly examined and tested by experts. Modern systems such as AES follow this principle.

Advantages: Symmetric encryption is generally fast, efficient and suitable for encrypting large amounts of data. **Limitation:** Securely distributing and managing the shared secret key is difficult, especially when many users communicate.

Q2. Distinguish between stream ciphers and block ciphers. Discuss confusion and diffusion, and explain ECB, CBC, CFB, OFB and CTR modes with strengths and weaknesses. [10 Marks]

Basis	Stream cipher	Block cipher
Processing	Encrypts data one bit or byte at a time.	Encrypts fixed-size blocks of data, such as 128-bit blocks.
Operation	Combines plaintext with a generated keystream, commonly using XOR.	Applies a keyed transformation to each block, often through a selected mode.
Speed/use	Useful for continuous or real-time data streams.	Useful for files, disks, databases and network messages.
Examples	RC4 (historical; now insecure), modern stream designs include ChaCha20.	AES and DES.

Confusion and diffusion

• **Confusion:** Makes the relationship between the secret key and ciphertext complex. Substitution operations, such as S-boxes, help provide confusion.

• **Diffusion:** Spreads the influence of one plaintext bit across many ciphertext bits. A small change in plaintext should affect many ciphertext bits. Permutations and mixing operations help provide diffusion.

Modes of operation

A mode of operation explains how a block cipher is applied to messages longer than one block. The following descriptions assume a block cipher with block size n and, where required, an initialisation vector (IV) or nonce.

Mode	How it works	Strengths	Weaknesses
ECB (Electronic Codebook)	Each plaintext block is encrypted independently with the same key.	Simple; blocks can be processed in parallel; errors do not spread to other blocks.	Identical plaintext blocks produce identical ciphertext blocks, revealing patterns. Not recommended for general data encryption.
CBC (Cipher Block Chaining)	Each plaintext block is XORed with the previous ciphertext block before encryption. The first block uses an IV.	Hides repeated-block patterns when used correctly; widely understood.	Encryption is sequential; padding is needed for incomplete blocks; IV must be unpredictable. Padding-oracle errors can be dangerous.
CFB (Cipher Feedback)	The previous ciphertext (or IV) is encrypted and the result is XORed with plaintext to form ciphertext.	Can process smaller units; turns a block cipher into a self-synchronising stream-like mode.	Sequential; transmission errors can affect more than one segment; requires careful IV use.
OFB (Output Feedback)	Repeatedly encrypts an internal state to generate a keystream, which is XORed with plaintext.	Bit errors in ciphertext generally affect only corresponding plaintext bits; keystream can be generated in advance.	Nonce/IV must never be reused with the same key; does not self-synchronise after lost bits.
CTR (Counter)	Encrypts successive counter values and XORs the outputs with plaintext.	Fast; parallelisable; supports random access; no padding required.	Counter/nonce values must never repeat under the same key; does not by itself authenticate data.

Important: Encryption alone does not necessarily provide integrity or authenticity. In practice, authenticated encryption (for example, AES-GCM) is preferred where supported.

Q3. Trace the historical development of symmetric encryption algorithms. [10 Marks]

Symmetric encryption developed from simple manual techniques into complex computer algorithms. Important milestones are described below in chronological order.

Period / milestone	Development and importance
Ancient period – Caesar cipher	The Caesar cipher replaces each letter with another letter a fixed number of positions away in the alphabet. It is easy to use but can be broken by trying the small number of possible shifts.
World War II – Enigma machine	The Enigma was an electromechanical rotor machine used by Germany. Its changing substitutions created a large number of possible settings. Allied cryptanalysts, including teams at Bletchley Park, broke Enigma traffic and demonstrated the importance of systematic cryptanalysis.
1940s – Claude Shannon	Claude Shannon's work on information theory and “Communication Theory of Secrecy Systems” (1949) gave a mathematical foundation to cryptography. He explained ideas including confusion and diffusion and described perfect secrecy.
1970s – Feistel structure	The Feistel network divides a block into two halves and processes them through repeated rounds. It allows encryption and decryption to use a related structure, with round keys applied in reverse order for decryption.
1977 – DES	The Data Encryption Standard (DES) was adopted as a U.S. Federal standard in 1977. It uses 64-bit blocks, a 56-bit effective key and 16 Feistel rounds. It became widely used, but its key later became too short against brute-force search.
1990s – Triple DES and other algorithms	Triple DES (3DES) applies DES three times to improve security. IDEA, Blowfish and RC4 also became notable: IDEA and Blowfish are block ciphers, while RC4 is a stream cipher. RC4 is now considered insecure and should not be used.
1997–2001 – AES competition	NIST organised an open competition to select a replacement for DES. Five finalists were evaluated publicly. Rijndael, designed by Joan Daemen and Vincent Rijmen, was selected and became AES in 2001.

Overall development

The history shows a move from simple letter substitution to mathematically designed, computer-based algorithms. Public analysis and standardisation became important: algorithms are openly reviewed, while secret keys provide confidentiality. AES replaced DES as the modern standard because it offers stronger key sizes and efficient implementation.

Q4. Explain classical substitution ciphers with their working principle, strengths and weaknesses. Why is frequency analysis effective against some but not others? [10 Marks]

A substitution cipher replaces plaintext letters or groups of letters with other letters or symbols. The order of the message is usually retained, but the symbols are changed.

(a) Caesar cipher

Each letter is shifted by a fixed number of positions. For a shift of 3, A becomes D, B becomes E and so on. Example: HELLO becomes KHOOR. Encryption can be written as C = (P + k) mod 26, where k is the shift value.

• **Strength:** Very simple and quick to perform by hand.

• **Weakness:** Only 25 non-zero shifts are possible, so brute force is easy. Letter-frequency patterns remain visible.

(b) Monoalphabetic substitution cipher

Each plaintext letter is consistently replaced by a different ciphertext letter according to a fixed substitution alphabet. For example, one mapping may replace A with Q, B with W and C with E. The same plaintext letter always maps to the same ciphertext letter.

• **Strength:** Has many possible substitution alphabets and is harder to solve by trying every key than the Caesar cipher.

• **Weakness:** Single-letter and repeated-letter patterns remain. Frequency analysis can often reveal common letters and words.

(c) Playfair cipher

Playfair is a digraph substitution cipher: it encrypts pairs of letters rather than single letters. A 5 × 5 matrix is formed using a keyword, with repeated letters removed and I/J usually combined. Plaintext is divided into pairs. If both letters are in the same row, each is replaced by the letter to its right; if in the same column, each is replaced by the letter below; otherwise, each is replaced by the letter in the same row and the other letter's column. A filler such as X is inserted when needed.

• **Strength:** Hides ordinary single-letter frequencies better than simple monoalphabetic substitution.

• **Weakness:** Digraph patterns and language statistics remain, so sufficient ciphertext can be analysed.

(d) Vigenere polyalphabetic cipher

The Vigenere cipher uses a repeating keyword to apply different Caesar shifts to successive letters. For example, the key determines the shift for each position. In a standard 26-letter alphabet, C = (P + K) mod 26. The same plaintext letter may therefore encrypt to different ciphertext letters at different positions.

• **Strength:** A longer, non-obvious key reduces the simple single-letter frequency pattern.

• **Weakness:** If the key repeats, its length may be estimated using methods such as Kasiski examination or the index of coincidence. The text can then be analysed as several Caesar ciphers.

Frequency analysis

In a natural language such as English, some letters occur more frequently than others (for example, E and T). Caesar and monoalphabetic substitution preserve these frequency patterns, so attackers can compare ciphertext frequencies with normal language frequencies. Playfair works on pairs and therefore disrupts single-letter frequencies, making basic analysis harder. Vigenere uses multiple alphabets, which also hides simple frequencies; however, a repeating key creates patterns that can be exploited. None of these classical methods is secure by modern standards.

Q5. Describe classical transposition techniques, product ciphers and the one-time pad. [8 Marks]

Transposition ciphers rearrange the positions of plaintext characters without changing the characters themselves. Unlike substitution, the letters remain the same but appear in a different order.

1. Rail fence cipher

The plaintext is written diagonally across a selected number of rows (rails), moving down and up in a zigzag pattern. The ciphertext is produced by reading the rows one by one. For example, with two rails, the letters are placed alternately on the first and second rows and then read row-wise. More rails create a different rearrangement.

• **Strength:** Simple and demonstrates the basic idea of rearranging characters.

• **Weakness:** The number of rails is limited and can be tried; letter frequencies are unchanged.

2. Columnar transposition cipher

The plaintext is written row by row under a keyword. The columns are then read in the order determined by the alphabetical order of the keyword's letters. The result is the ciphertext. The receiver uses the same keyword to restore the column order and read the message row by row.

• **Strength:** More complex rearrangement than a basic rail fence cipher.

• **Weakness:** Letter frequencies are unchanged, and repeated patterns or key-length clues can assist cryptanalysis.

3. Product ciphers

A product cipher combines substitution and transposition in multiple stages. Substitution changes the identity of symbols, while transposition changes their positions. Repeating these operations creates confusion and diffusion. Modern block ciphers, including DES and AES, use repeated rounds of substitution, permutation or mixing, and key addition. Their designs are much more sophisticated than classical hand ciphers.

4. One-time pad (OTP)

A one-time pad combines the plaintext with a truly random key that is exactly as long as the message. The key is used only once and kept completely secret. For binary data, encryption is commonly performed using XOR: C = P XOR K. Decryption uses P = C XOR K.

Why it is theoretically unbreakable: If the key is truly random, as long as the message, secret and never reused, every possible plaintext is equally consistent with a given ciphertext. This provides perfect secrecy.

Why it is impractical: Generating and securely sharing a random key as long as every message is difficult. The key must be stored safely, never reused, and securely destroyed after use. Key distribution and management become costly for frequent communication.

Q6. Describe the structure of DES. Explain the Feistel structure, block size and key size, permutations, 16 rounds, round function and key schedule. [12 Marks]

The Data Encryption Standard (DES) is a symmetric block cipher based on a 16-round Feistel network. It was adopted as a U.S. standard in 1977. DES is now considered insecure for protecting new information because its effective key is too short.

Basic specifications

• **Block size:** 64 bits of plaintext are processed at a time.

• **Key size:** The supplied key is 64 bits, but 8 bits are used for parity, leaving an effective key length of 56 bits.

• **Rounds:** 16 Feistel rounds.

• **Structure:** The block is divided into two 32-bit halves, called L and R.

Overall DES process

64-bit plaintext	Initial permutation (IP)	Split into L0 and R0
16 Feistel rounds (round keys K1–K16)	Swap final halves	Inverse initial permutation (IP■[1])

The result is a 64-bit ciphertext. The initial permutation (IP) rearranges the input bits according to a fixed table. The final permutation is the inverse of IP and restores the bit positions after the rounds.

Feistel round structure

In each round i, the left and right halves are updated as follows: **L■ = R■■■** and **R■ = L■■■ XOR F(R■■■, K■)**. Here, F is the DES round function and K■ is the 48-bit round key. The same structure supports decryption by applying the round keys in reverse order.

R (32 bits)	Expansion E 32 → 48 bits	XOR with K■ (48 bits)	S-boxes 48 → 32 bits	P permutation (32 bits)

The output of F is XORed with the previous left half to form the new right half; the previous right half becomes the new left half.

Steps inside the round function F

• **Expansion:** The 32-bit right half is expanded to 48 bits using the expansion permutation E. Some bits are repeated.

• **Key mixing:** The expanded 48-bit value is XORed with the 48-bit round key.

• **S-box substitution:** The 48-bit result is divided into eight 6-bit groups. Each group passes through an S-box and becomes 4 bits, producing 32 bits. S-boxes provide non-linearity.

• **P-box permutation:** The 32 bits are rearranged using a fixed permutation P to spread their influence.

DES key schedule

The 56 effective key bits are first rearranged by Permuted Choice 1 (PC-1) and split into two 28-bit halves, C and D. In each round, the halves are circularly shifted left by one or two positions according to the DES shift schedule. Permuted Choice 2 (PC-2) selects 48 bits from the shifted halves to create the round key. This process generates sixteen 48-bit round keys.

Note: DES has historical importance, but its 56-bit key can be searched by modern hardware. AES is preferred for current security needs.

Q7. Discuss the strength and weaknesses of DES. Explain Double DES and Triple DES and the meet-in-the-middle attack. [10 Marks]

DES introduced a carefully designed round structure and became an important standard. However, improvements in computing power made its 56-bit key inadequate.

Strengths of DES

• **Avalanche effect:** A small change in the plaintext or key (such as flipping one bit) should change many ciphertext bits, ideally about half. This makes relationships between input and output difficult to predict.

• **Well-studied design:** DES received extensive public analysis and helped establish modern block-cipher design practices.

• **Efficient hardware implementation:** DES was practical and fast in hardware when introduced.

Weaknesses of DES

• **Short key:** Its effective key length is only 56 bits, giving 2■■ possible keys. This is vulnerable to exhaustive brute-force search with sufficiently powerful hardware.

• **Brute-force attack:** An attacker tries possible keys until the decrypted output looks like valid plaintext. The short key is the main practical weakness.

• **Differential cryptanalysis:** A method that studies how differences in selected plaintext pairs affect ciphertext differences. DES was designed to resist many such attacks, but the method is important in analysing block ciphers.

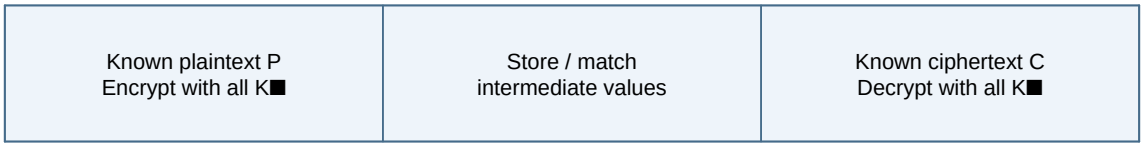
• **Linear cryptanalysis:** Uses statistical relationships among plaintext bits, ciphertext bits and key bits. It can reduce the work needed compared with a complete key search under suitable conditions.

Double DES

Double DES applies DES encryption twice with two independent 56-bit keys: **C = E(K■, E(K■, P))**. It appears to provide a 112-bit key space. However, it is vulnerable to a meet-in-the-middle attack, so its security is not equivalent to a straightforward 112-bit key.

Meet-in-the-middle attack

The attacker has a known plaintext P and its corresponding ciphertext C. The attacker encrypts P using every possible K■ and stores the intermediate results. Separately, the attacker decrypts C using every possible K■. Matching intermediate values can reveal candidate key pairs. This reduces the work substantially compared with trying every pair of keys directly, although substantial storage and verification are required.



Triple DES (3DES)

Triple DES applies DES three times, commonly using the Encrypt–Decrypt–Encrypt (EDE) sequence: **C = E(K■, D(K■, E(K■, P)))**. EDE allows compatibility with single DES in a special equal-key configuration.

• **Two-key 3DES:** Uses K■ and K■, with K■ = K■. Its nominal key material is 112 bits.

• **Three-key 3DES:** Uses three independent 56-bit DES keys, for 168 bits of key material.

3DES improved resistance to brute-force attacks compared with single DES, but it is much slower than modern AES and retains DES's 64-bit block size. It is now deprecated for most new applications. AES is the preferred modern replacement.

Q8. Explain AES. Describe its block size, key lengths, rounds, transformations, key expansion, decryption and comparison with DES. [10 Marks]

The Advanced Encryption Standard (AES) is a widely used symmetric block cipher selected by the U.S. National Institute of Standards and Technology (NIST). It is based on the Rijndael algorithm and was adopted in 2001. AES is designed to provide strong security and efficient software and hardware performance.

Main specifications

Feature	AES details
Block size	Fixed at 128 bits (16 bytes).
Key lengths	128, 192 and 256 bits.
Number of rounds	10 rounds for 128-bit key; 12 rounds for 192-bit key; 14 rounds for 256-bit key.
Structure	Substitution–permutation network (not a Feistel network).

AES encryption process

The 128-bit input is arranged as a 4 × 4 byte matrix called the state. The initial step is AddRoundKey. Then the main rounds perform SubBytes, ShiftRows, MixColumns and AddRoundKey. In the final round, MixColumns is omitted.

Four round transformations

• **1. SubBytes:** Each byte in the state is replaced using a non-linear substitution table called the S-box. It provides confusion.

• **2. ShiftRows:** The rows of the state are cyclically shifted by different offsets. This moves bytes to new positions and helps spread information.

• **3. MixColumns:** Each column is mathematically mixed so that the bytes influence one another. This provides diffusion. It is not performed in the final encryption round.

• **4. AddRoundKey:** The state is combined with a round key using bitwise XOR. Each round key is derived from the original key.

Key expansion

AES uses a key expansion algorithm to generate a separate round key for each round. The original key is expanded into a key schedule. The process uses operations such as RotWord (rotating a word), SubWord (applying the S-box), and round constants (Rcon). The number of generated round-key words depends on whether the original key is 128, 192 or 256 bits.

AES decryption

Decryption reverses the encryption process using the expanded round keys in reverse order. It applies the inverse transformations: InvShiftRows, InvSubBytes, AddRoundKey and InvMixColumns. The order is arranged so that the original plaintext is recovered. AES decryption is not simply the same operations in the same order; it uses the corresponding inverse transformations.

AES and DES comparison

Basis	DES	AES
Block size	64 bits	128 bits
Key size	56 effective bits	128, 192 or 256 bits
Structure	16-round Feistel network	Substitution–permutation network
Rounds	16	10, 12 or 14 depending on key size
Security	Insecure for modern protection because of short key	Strong when correctly implemented and used
Performance	Efficient in older hardware; 3DES is slow	Efficient in modern software and hardware, including AES acceleration

Conclusion: AES provides a larger block size and stronger key options than DES. Its modern design and efficient implementation make it the standard choice for protecting data. DES is mainly of historical and educational importance.

References

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End of Assignment